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A contrastive learning and knowledge distillation-based framework for efficient federated intrusion detection in IoT

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ABSTRACT

In the era of pervasive connectivity, the widespread deployment of Internet of Things (IoT) devices across various applications has led to a rise in malicious attacks, necessitating the development of robust network intrusion detection systems (IDS) for IoT. Traditional deep learning-based IDS are challenged by the diversity of IoT devices, resource constraints, and the non-identically and independently distributed (non-IID) nature of IoT data. To address these issues, this paper introduces two solutions: CF-IDS and KF-IDS. CF-IDS is a federated learning-based IDS designed for cloud environments, preserving data privacy and addressing non-IID challenges. It utilizes contrastive learning and clustering to create pseudo-labels for public datasets, improving cold-start performance by aligning client models in the early training process. KF-IDS, as an additional (but optional) subsequent step of CF-IDS, based on knowledge distillation and mutual information theory, provides a off-line IDS solution by distilling CF-IDS into lite models. Through local deployment on low-computational-power end devices, KF-IDS avoiding upload raw data to the cloud. Experimental evaluations on public IoT datasets demonstrate that CF-IDS achieves 80.71% accuracy in non-IID scenarios, outperforming other federated learning methods such as FedNova. KF-IDS reduces time consumption by 46.81% and memory usage by 48.30% than CF-IDS while maintaining accuracy, which emphasizes the low cost of KF-IDS, such as deployed on affordable NVIDIA Jetson Nano rather than expensive RTX 4090 or Tesla A100.

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1. Introduction

With the rise of IoT technology, security challenges are becoming more evident. IoT devices, which serve as endpoints for data collection and transmission, often gather personal identity details (Madakam et al., 2015; Sarker et al., 2023), daily habits, and even sensitive health information to create tailored recommendation systems or data analysis reports that align with users' behaviors. However, when this data is uploaded to the cloud, practices like 'transparent data transmission' or the failure to adequately desensitize the data can lead to breaches of personal privacy and the risk of it being sold on the dark web (Tawalbeh et al., 2020). This presents significant challenges to data and privacy protection. Unfortunately, many IoT devices prioritize functionality and user-friendliness during design and deployment, often overlooking information security. Attackers can exploit these security gaps to threaten users' privacy and safety, potentially leading to large-scale data breaches. Thus, building a secure and reliable IoT ecosystem is crucial in today's era of data proliferation.

As a critical element of network security infrastructure, the intrusion detection system (IDS) in the IoT environment plays a crucial role in monitoring and protecting networks. These systems are essential for safeguarding IoT devices and their networks from unauthorized access and malicious activities by continuously analysing network traffic and detecting any unusual patterns that suggest potential intrusions. Intrusion detection systems can be categorized as either cloud-based or terminal-based, depending on where they are deployed, each offering distinct benefits. Cloud-based IoT intrusion detection systems are well-suited for large-scale (Alkadi et al., 2020; Madakam et al., 2015), centralized data analysis and processing, but they typically require significant bandwidth and resources. On the other hand, terminal-based detection systems eliminate the need for data transmission and instead use homomorphic encryption channels to upload or download trained model parameters (Eskandari et al., 2020), which are then aggregated to build an intrusion detection model through federated learning. However, the limited computing power of terminal

devices may struggle to support detection algorithms that rely on large, complex deep learning networks.

In recent years, whether it is cloud or terminal IoT intrusion detection systems, application scenarios based on deep learning technology have been widely developed, and have gradually become a research topic in security fields such as host anomalies and network risk situation awareness and prevention. Compared with traditional detection solutions based on statistics, pattern matching and other methods, deep learning has demonstrated superior performance in various complex pattern recognition tasks with its end-to-end paradigm and efficient feature extraction capabilities. To improve the accuracy and efficiency of intrusion detection. For example, X. Li et al. (2020) proposed an autoencoder intrusion detection method based on the random forest algorithm, which uses a shallow autoencoder neural network to reduce computational complexity and improve accuracy. In addition, Naseer et al. (2018) proposed a deep learning intrusion detection system, using convolutional neural networks, autoencoders, and recurrent neural networks to train and test on the NSL-KDD dataset, where NSL-KDD dataset is an enhanced version of the original KDD Cup 1999 dataset, designed to serve as a benchmark for evaluating intrusion detection systems by addressing issues such as redundant records and unbalanced attack type distributions. It offers a refined collection of network connection records, each annotated with labels indicating normal activity or specific types of attacks, thereby supporting more accurate and reliable performance assessments of cybersecurity mechanisms.

While deep learning-based intrusion detection systems (IDS) have demonstrated remarkable efficacy across various domains, their direct application within IoT environments presents distinct challenges. These include privacy concerns regarding IoT data, the non-IID nature of data distributions, the extensive diversity in IoT device ecosystems, and the inherent limitations of device resources. Such factors critically impact both the training and inference phases of IDS models. In particular, privacy issues and the non-IID characteristics of the datasets introduce unforeseen complexities in model training. Furthermore, the heterogeneity of IoT devices and their resource constraints complicate the inference process for endpoint security solutions.

To this non-IID challenge, various federated learning-based IDS methods has proposed. However, these IDS methods still relies on manually defined labels and is supervised learning. In addition to this, the highly non-IID nature of IoT data, where each participant may be from a different type of device such as a router or a physical firewall, leads to the fact that the federated aggregation mechanism on which federation learning relies

may perform poorly in the initial phase of training (i.e. cold starting phase), requiring careful tuning of the hyperparameters or longer training time to converge to the desired state.

To address these remaining challenges, we propose an innovative semi-supervised federated learning approach designed to enhance performance by leveraging prior knowledge. Our methodology, termed CF-IDS, employs contrastive learning on a central server to generate pseudo-labels from public datasets before initiating client-side training. This strategy facilitates alignment of initial knowledge across clients, mitigating conflicts and expediting the cold-start phase. Additionally, our KF-IDS integrates game-theoretic Shapley values into the voting mechanism during the aggregation phase. By quantifying each client's contribution based on historical data, this method measures the impact of individual predictions on the overall loss function, thereby replacing the traditional one-person-one-vote system with a weighted voting scheme. This approach not only accounts for collaborative relationships among participants but also optimizes outcomes even under conditions resembling the prisoner's dilemma.

In comparison with existing federated learning-based IDS methodologies, CF-IDS uniquely leverages contrastive learning to tackle the non-IID data challenge, significantly improving the initialization phase and enhancing model convergence rates. Meanwhile, KF-IDS introduces a novel weighting mechanism through Shapley values, which has been empirically validated to provide superior accuracy and robustness in federated learning settings. These innovations collectively represent a substantial advancement in the field, addressing critical limitations of current approaches and paving the way for more effective, scalable, and secure intrusion detection systems tailored for the diverse landscape of IoT.

The main contributions of this article can be summarized as follows:

- (1) *Design of CF-IDS*: A novel semi-supervised federated learning approach utilizing contrastive learning on a central server to generate pseudo-labels, which enhances the initial training phase and addresses the non-IID data challenge in IoT intrusion detection.
- (2) *Development of KF-IDS*: An lite federated learning framework that integrates game-theoretic Shapley values into the aggregation process, enabling a weighted voting mechanism based on each client's contribution to improve accuracy and robustness.
- (3) *Significant advancement*: The proposed CF-IDS and KF-IDS collectively address key limitations of existing methods, particularly the non-IID data problem

and the cold-start phase, leading to more effective, affordable, and secure intrusion detection systems tailored for the diverse IoT environment.

2. Related works

2.1. Deep learning and advanced IDS frameworks

In recent years, deep learning methodologies have garnered considerable attention within the realm of intrusion detection system (IDS) design. This prominence is largely attributed to their exceptional capability in discerning patterns even amidst complex and obfuscated datasets (Mirsky et al., 2018). For example, an autoencoder-based system was introduced to calculate a final anomaly score to quantify the extent of anomalies present. In Mirsky et al. (2018), the researchers used the Bot-IoT dataset for validation. The dataset contains four main attack types: distributed denial-of-service attacks (DDoS), denial-of-service attacks (DoS), scanning (Scan), and data theft (Theft). In addition, unknown attack categories were considered. Meanwhile, Bovenzi et al. (2020) proposed a two-phase IDS framework: the initial phase employs a multimodal approach for anomaly detection, followed by a second phase dedicated to classifying the nature of detected attacks. Further advancements include utilizing deep belief networks to scrutinize harmful behaviors within network environments (Balakrishnan et al., 2021), and developing ensemble models specifically for detecting SCADA-specific threats in traffic networks (Huda et al., 2018).

2.2. Addressing challenges in deep learning models and federated learning

Despite these innovations, a notable challenge encountered by these models was the risk of gradient vanishing, particularly when using sigmoid activation functions within deep belief networks during backpropagation. To address this issue, hybrid methods integrating convolutional neural networks (CNN) with weighted long short-term memory (WDLSTM) networks were introduced to identify common cyberattack categories more effectively (Hassan et al., 2020). Moreover, federated learning has emerged as a promising solution to alleviate risks concerning data privacy breaches associated with centralized machine learning frameworks (Cheng et al., 2022; Kang et al., 2019; Popoola et al., 2021). Within a federated learning framework, devices collaboratively train local models by periodically sharing and aggregating model parameters or gradients on a central server without uploading raw data, thereby preserving data privacy (Agrawal et al., 2022).

Specific in IDS context, Mahmud et al. (2024) introduces a Privacy-Preserving Federated Learning-based Intrusion Detection Technique designed for Cyber-Physical Systems, which ensures sensitive information remains within individual IoT devices during model training by utilizing the Federated Averaging (FedAvg) algorithm to iteratively refine a global model. Javeed et al. (2024) proposes a hybrid Federated Learning (FL) based Intrusion Detection System (IDS) that utilizes CNN-BiLSTM models for identifying and classifying attack types in datasets such as CICIDS 2017 and Edge-IIoTset, achieving high accuracy rates across multiple performance metrics. Hajj et al. (2023) proposes a novel lightweight intrusion detection system for IoT networks that utilizes federated learning and cluster-based sampling to enhance detection accuracy while preserving data privacy and minimizing resource consumption. Pei et al. (2023) proposes FedMaIDE, a semi-supervised federated learning framework for IoT malware detection that addresses data privacy concerns and the challenges of data scarcity and labeling in traditional centralized approaches. Specifically, FedMaIDE utilizes a knowledge transfer mechanism to automatically label unlabeled data, allowing users' private data to be used for training without exposing sensitive information.

2.3. Federated learning applications and proposed approach

Among federated learning-based IDS tailored for IoT, pioneers like Kang et al. (2019) introduced the concept of reputation as a metric for assessing the reliability and trustworthiness of mobile devices, ensuring more dependable federated learning processes. Furthermore, semi-supervised systems grounded in Federated Learning (FL) aim at addressing privacy concerns (Zhao et al., 2022). In the domain of Semi-Supervised Federated Learning (SSFL), researchers have devised methods to partition model parameters between labeled servers and unlabeled clients (Jeong et al., 2021), alongside strategies that involve concurrent training and aggregation of model parameters with group-wise reweighting (Zhang et al., 2021). In the context of IoT-oriented IDS, Aouedi et al. (2023) leverages both labeled and unlabeled data while ensuring data privacy by training local autoencoders on each device and aggregating them into a global model using federated learning. The global model is then further trained using a small amount of labeled data, achieving high classification performance with minimal communication overhead and privacy concerns. Naeem et al. (2023) proposes a novel IDS in Zero Touch Network and Service Management (ZSM) networks. The FL-SSAL framework

addresses the challenges of limited labeled data and privacy concerns in 6G networks by combining federated learning, semi-supervised learning, and active learning techniques.

2.4. Our propose

In this paper, we propose a novel semi-supervised federated learning approach, CF-IDS, which utilizes contrastive learning for pseudo-label generation; and a lite solution, KF-IDS, based on Mutual Information-based Knowledge Distillation, aiming to bridge the existing gaps within FL-based intrusion detection systems designed for IoT scenarios, specifically by addressing the challenges posed by the non-IID nature of IoT data, the reliance on supervised learning with manually defined labels, and the poor performance during the initial training phase (cold-start problem). We introduced the core methodology of CF-IDS in our previous work (Ma et al., 2024), and the newly proposed KF-IDS builds upon and extends that earlier work.

3. Preliminary

3.1. Contrastive learning

Contrastive learning constitutes a pivotal paradigm within the realm of machine learning, especially prominent in unsupervised and self-supervised learning sectors. This methodology entails educating a model to discern between similar and dissimilar data instances through the juxtaposition of positive and negative samples. Such an approach is fundamentally anchored in the notion that exposing the model to both contrasting and complementary information facilitates the acquisition of robust representations.

The essence of contrastive learning is to cultivate a feature space wherein similar instances are positioned closely, whereas dissimilar ones are distanced apart. This objective is actualized via employing a loss function, for instance, the contrastive loss, which quantifies the discrepancy between the embedded representations of positive and negative samples. Minimization of this loss drives the model to cluster similar instances closely together whilst distancing dissimilar ones.

The strength of contrastive learning emanates from its capacity to harness vast quantities of unlabeled data, a trait particularly advantageous in contexts where labeled data is limited or costly to procure. Furthermore, contrastive learning has demonstrated an enhancement in models' generalization abilities as well as their resilience against noise and adversarial assaults.

3.2. K-Means clustering

K-Means Clustering is a popular method of unsupervised machine learning whose primary goal is to partition n observations into k clusters, where each observation belongs to the cluster with the nearest mean, serving as a prototype of the cluster. This method requires the number of clusters k to be specified beforehand. The algorithm works through an iterative process that aims to minimize the within-cluster variance or squared Euclidean distance between the data points and their corresponding cluster centroids.

The steps in K-Means clustering are as follows:

- *Initialization*: Randomly select k points (data points) from the dataset as initial centroids.
- *Assignment*: Assign each data point to the nearest centroid. The 'nearest' is typically measured using Euclidean distance, though other metrics can also be used.
- *Update*: Recalculate the centroids by taking the average of all points assigned to each cluster.
- *Repeat*: Repeat the assignment and update steps until either the centroids do not change significantly between consecutive iterations, indicating convergence, or a maximum number of iterations is reached.

K-Means is widely used due to its simplicity and speed, making it efficient for clustering large datasets. However, it has limitations such as sensitivity to the initial placement of centroids and the need to specify k in advance. It also tends to find clusters of relatively uniform size and does not perform well with datasets having clusters of varying sizes and densities. Despite these limitations, K-Means remains a fundamental tool in exploratory data analysis, image segmentation, customer segmentation, and many other applications.

3.3. Mutual information-based knowledge distillation

Mutual Information-based Knowledge Distillation is an advanced technique used in the field of machine learning, particularly aimed at improving the efficiency and performance of neural networks through the process of knowledge distillation.

Knowledge distillation involves training a smaller, more efficient 'student' model to mimic the behavior of a larger, more complex 'teacher' model, with the goal of retaining the teacher's accuracy while reducing computational requirements.

In the context of mutual information, this approach leverages the concept of mutual information—a measure

from information theory that quantifies the amount of information obtained about one random variable through another—to guide the distillation process. Mutual information measures the dependency between the variables, making it a powerful tool for capturing the relationships within data and between layers in neural networks. Mathematically, the mutual information $I(X; Y)$ between two random variables X and Y is defined as:

$$I(X; Y) = \sum_{x \in X} \sum_{y \in Y} p(x, y) \log \frac{p(x, y)}{p(x)p(y)} \quad (1)$$

When applied to knowledge distillation, mutual information helps to identify and transfer the most informative aspects of the teacher model's knowledge to the student model. Specifically, it focuses on aligning the distributions of representations learned by the teacher and student models, ensuring that the student captures not just the outputs but also the nuanced understanding of the data that the teacher has developed. Mathematically, the goal of mutual information-based knowledge distillation is:

$$\max \sum_{x \in X} \sum_{y \in Y} I(T(x), S(x)) \quad (2)$$

where $T(x), S(x)$ represents the output (or intermediate features) of the teacher model and the student model given input x .

By maximizing the mutual information between the teacher and student models, or equivalently minimizing the loss based on mutual information, the student model can learn to generalize better, often achieving improved performance over traditional distillation methods. This approach can lead to more effective compression of large models into smaller ones, making it highly valuable for deploying machine learning models in resource-constrained environments.

4. Methodology

In this paper, we present two IDS solutions: the CF-IDS and the KF-IDS. where the KF-IDS is a lightweight distilled version of the CF-IDS that allows local deployment on low-computing-power IoT devices. These two solutions are described in detail in the next two subsections.

4.1. The proposed CF-IDS

The proposed IoT intrusion detection system leverages federated learning within an IoT environment comprising multiple clients and a centralized server. This system uses contrastive learning to generate embeddings so that to be clustered and produce pseudo labels, which are then utilized to train client-based classifiers and discriminators,

as well as to aggregate the global model at the central server. The system architecture is showed in Figure 1.

Before the federated training begins, the central server accesses an open dataset of unlabeled traffic data. It uses contrastive learning to pre-embed this data and applies clustering methods to assign pseudo labels. These pseudo labels are then distributed to the clients for use in their local training.

During the federated training stage, each client trains a local classifier model using its private dataset. Since the local classifier may not be familiar with certain classes of traffic not represented in the private dataset, it can misclassify unfamiliar unlabeled traffic packets. To address this, a discriminator is introduced to determine whether an unlabeled traffic packet is 'familiar' or not. Pseudo labels are assigned to unfamiliar traffic, improving the quality of the predicted soft labels. Both the classifier and discriminator networks are based on the Vision Transformer (ViT) model (Khan et al., 2022), which is designed to effectively extract deep features from traffic packets.

To minimize communication costs during the federated learning process, clients rely on the assigned pseudo labels for training. The exchange cycle between the central server and clients continues iteratively until the global model converges, which is determined by monitoring performance metrics such as classification accuracy and prediction consistency across clients. Once the central server gathers all predictions from clients, it employs a voting mechanism to determine the global classification labels for the unlabeled public data rather than directly aggregating the results. Finally, the intrusion is detected based on these globally determined labels, and the central server makes the final decision regarding the presence of a security threat. These aggregated labels are then sent back to the clients for further classifier training in subsequent stages.

4.1.1. Pseudo labels generation

We suppose that the centralized server has collected a batch of open dataset $D^{\text{open}} = x_i^{\text{open}}$, now the centralized server provides the pseudo labels for this open dataset, the pseudo labels are generated by a ViT model and a K-means cluster model. The ViT model will get trained through contrastive learning strategy, and generates the embedding for the open dataset. These embedding will then be clustered by the K-nearest neighbor algorithm, and the pseudo labels will finally be generated based on the clustering results. The specific training process of ViT model is as follows.

First, two random augmentation processes are deployed to produce two different views D_1^{open} and D_2^{open} from an original dataset D^{open} . These augmentations may include various transformations such as cropping,

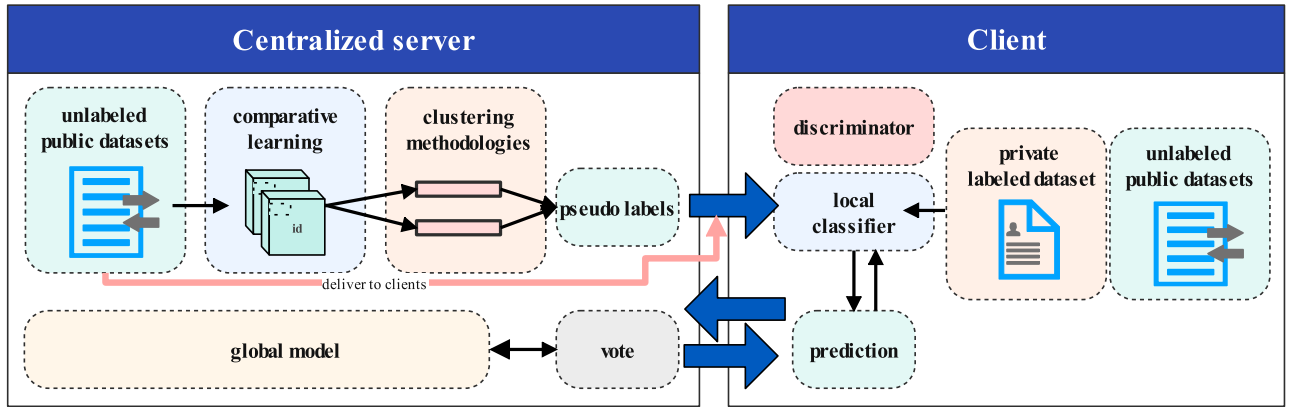


Figure 1. The proposed CF-IDS architecture.

color jittering, or other image alterations that ensure the semantic information remains intact but introduces variability in the input. This step is crucial for simulating diverse real-world scenarios and enhancing the model's robustness.

Subsequently, these augmented views are fed into a Vision Transformer (ViT) model, wherein the weights are shared across both inputs. The ViT model is tasked with extracting high-level feature representations, resulting in two distinct node embedding matrices. This process leverages the self-attention mechanism inherent in transformers, enabling the model to capture intricate patterns and dependencies within the data.

Following the extraction of embeddings, a comparative loss function, also known as contrastive loss, is applied to these matrices. The primary objective of this loss function is to bring closer the embeddings of positive node pairs while pushing apart those of negative node pairs in the latent space. Positive node pairs are defined as the embeddings of the same node across the two different views D_1^{open} and D_2^{open} , whereas negative node pairs consist of embeddings pertaining to different nodes either within the same view or across the different views.

The contrastive loss for a example i in the first view can be specifically formulated as:

$$L(e_i^1, e_i^2) = -\log \frac{e^{s(e_i^1, e_i^2)/\tau}}{e^{s(e_i^1, e_i^2)/\tau} + \sum_{j \neq i} (e^{s(e_i^1, e_j^1)/\tau} + e^{s(e_i^1, e_j^2)/\tau})} \quad (3)$$

where e_i^1, e_i^2 are the embeddings of x_i^{open} in the views of D_1^{open} and D_2^{open} . $s(\cdot)$ is the similarity measure function, usually cosine similarity. And τ is a hyperparameter. We calculate the losses of the above two views, and get the

entire loss as follows:

$$L_{\text{con}}(e_i^1, e_i^2) = \sum_i^V (L(e_i^1, e_i^2) + L(e_i^2, e_i^1)) \quad (4)$$

This formulation ensures that the model learns to distinguish between positive and negative pairs effectively, thereby improving its discriminative power. Through iterative training with this contrastive loss, the model gradually refines its ability to map semantically similar nodes close to each other in the embedding space, which is pivotal for tasks such as image classification.

After we optimize ViT by minimizing $L_{\text{con}}(e_i^1, e_i^2)$, we believe the optimized ViT model has good feature extraction capabilities, So we mark ViT's embedding of x_i^{open} as e_i^{open} .

Next we cluster the embeddings e_i^{open} . First, K samples are randomly selected as the initial clustering centers, expressed as $\{c_1^0, c_1^1, \dots, c_1^K\}$. For each sample e_i^{open} , calculate the distance between it and each cluster center $\{c_1^0, c_1^1, \dots, c_1^K\}$, and assign it to the cluster corresponding to the Euclidean distance with the nearest cluster center. The probability that sample e_i^{open} is assigned to the k th cluster is expressed as:

$$r_{ik} = \begin{cases} 1, & \text{if } k = \arg \min_j \|e_i^{\text{open}} - c_j\|^2 \\ 0, & \text{else} \end{cases} \quad (5)$$

For each cluster, the cluster center is updated to be the mean of all samples in the cluster. The new cluster center is expressed as:

$$c_k = \frac{\sum_{i=1}^n r_{ik} e_i^{\text{open}}}{\sum_{i=1}^n r_{ik}} \quad (6)$$

The calculation process for r_{ik} and c_k are performed iteratively until the cluster center converges or the maximum number of iterations is reached. Finally, we calculate the distance between each e_i^{open} and all cluster centers, and regard the center c_i^{open} of the nearest cluster as the pseudo label of e_i^{open} (i.e. x_i^{open}).

4.1.2. Training the client classifier

We assume that there are k clients (i.e. IoT nodes), and each client has collected a batch of labeled data. We record the private data on the k th client as $D_k = \{x_i^k, y_i^k\}$. First, the centralized server issues the open dataset with pseudo labels and corresponding labels $D^{\text{open}} = \{x_i^{\text{open}}, y_i^{\text{open}}\}$ to each client before starting training. Then, each client initializes as classifier model f_k^c and f_k^d . Next, the client trains f_k^c according to $D_k \cup D^{\text{open}}$, and calculates the cross entropy of the loss of x_i^{open} and y_i^{open} as follows:

$$L_{\text{training}} = - \sum_{c=1}^{|F|} y_{i,c}^k \ln f_k^c(x_i^k) \quad (7)$$

Where $|F|$ is the number of clusters. In the IoT intrusion detection system, the default $|F| = 2$. The prediction label for x_i^k is noted as $y_{i,c}^k = \arg \max(f_k^c(x_i^k))$.

4.1.3. Training the client discriminator

We first create two empty sets $D_{\text{familiar}}^k = D_{\text{unfamiliar}}^k = \emptyset$ for each client. For each $x_i^{\text{open}} \in D^{\text{open}}$, we acquire the $f_k^c(x_i^{\text{open}})$ for computing the maximum confidence $c_k(x_i^{\text{open}})$. When $c_k(x_i^{\text{open}}) < \theta$, we believe that x_i^{open} will be recognized as 'unfamiliar' by the k th client, and add x_i^{open} to $D_{\text{unfamiliar}}^k$. Otherwise, we add it to D_{familiar}^k .

Next, we apply $D_{\text{familiar}}^k \cup D_{\text{unfamiliar}}^k$ to train f_k^d , whose labels are $l(x_i^{\text{open}} \in D_{\text{familiar}}^k)$, where $l(\text{ture}) = 1$ and $l(\text{false}) = 0$. And we decide the upload information sending to the centralized server by the f_k^d 's prediction label $y_{i,c}^{(k,\text{open})'} = \arg \max(f_k^d(x_i^{\text{open}}))$.

Note that we also ask each client k to upload the loss value of $y_{i,c}$, denoted as $L_{\text{training}}^k[y_{i,c}]$. Once the predictions of k clients are collected, the centralized server initiates a Shapley-based voting process using the uploaded prediction labels. $\{y_{i,c}^{(1,\text{open})'}, y_{i,c}^{(2,\text{open})'}, \dots, y_{i,c}^{(K,\text{open})'}\}$ from all clients to determine the global label $y_{i,c}^{(\text{glo},\text{open})'}$.

Specifically, for the training epoch i , and the chosen client k with their predict $y_{i,c}$, we denote the value function as:

$$v(k) = \sum_i L_{\text{training}}^k[y_{i,c}] \quad (8)$$

We can then apply the formula for the Shapley value to calculate the Shapley value $\Phi(k)$ for each voter. Assuming a total of K voters, the formula for the Shapley value can be rewritten as:

$$\Phi(k) = \sum_{S \subseteq N \setminus \{k\}} \frac{|S|! \cdot (n - |S| - 1)!}{K!} \cdot \left(\sum_{j \in S \cup \{k\}} v(k) - \sum_{j \in S} v(k) \right) \quad (9)$$

where N stand for the all client, and $|N| = K$.

We only upload the x_i^{open} that prediction label $y_{i,c}^{(k,\text{open})'} = 1$, then the centralized server executes a voting process for the upload prediction labels $\{y_{i,c}^{(1,\text{open})'}, y_{i,c}^{(2,\text{open})'}, \dots, y_{i,c}^{(K,\text{open})'}\}$ from all the clients to select the $y_{i,c}^{(\text{glo},\text{open})'}$, where Mode is the mode function.

$$y_{i,c}^{(\text{glo},\text{open})'} = \text{Mode} \left(\{y_{i,c}^{(1,\text{open})'}, y_{i,c}^{(2,\text{open})'}, \dots, y_{i,c}^{(K,\text{open})'}\} \right) \quad (10)$$

After obtaining the $y_{i,c}^{(\text{glo},\text{open})'}$, the centralized server issues the $y_{i,c}^{(\text{glo},\text{open})'}$ to each f_k^c , and continues to train the f_k^c by calculating the loss function as follows:

$$L_{\text{training}} = - \sum_{c=1}^{|F|} y_{i,c}^k \ln f_k^c(x_i^k) - \beta \sum_{c=1}^{|F|} y_{i,c}^{(\text{glo},\text{open})'} \ln f_k^c(x_i^{\text{open}}) \quad (11)$$

where β is a hyper-parameter to balance the importance of two loss function terms. With the iteration of the federated learning model, the importance of model training is increased for local private dataset, and the loss function of pseudo labels will gradually enter a decreased trend. Supposed the local model weight $\alpha = 1$, the formula of the balancing parameter is as follows:

$$\beta = \alpha \frac{1}{e^{T_{\text{current}}}} \left(1 + \cos \left(\frac{T_{\text{current}}}{T_{\text{total}}} \pi \right) \right) \quad (12)$$

where T_{current} represents the current number of training times, T_{total} represents the total number of training times, and α is the hyperparameter.

4.1.4. Aggregating the global model

The training of the global model relies on $y_{i,c}^{(\text{glo},\text{open})'}$ and c_i^{open} . In the initial rounds, the training of the global model is mainly based on c_i^{open} , and in the subsequent rounds is mainly based on $y_{i,c}^{(\text{glo},\text{open})'}$. This process still requires a balancing parameter β to balance $y_{i,c}^{(\text{glo},\text{open})'}$ and c_i^{open} . The training loss of the global model $L_{\text{training}}^{\text{glo}}$ is as follows:

$$L_{\text{training}}^{\text{glo}} = - \sum_{c=1}^{|F|} y_{i,c}^k \ln f_k^c(x_i^k) - \beta \sum_{c=1}^{|F|} y_{i,c}^{(\text{glo},\text{open})'} \ln f_k^c(x_i^{\text{open}}) \quad (13)$$

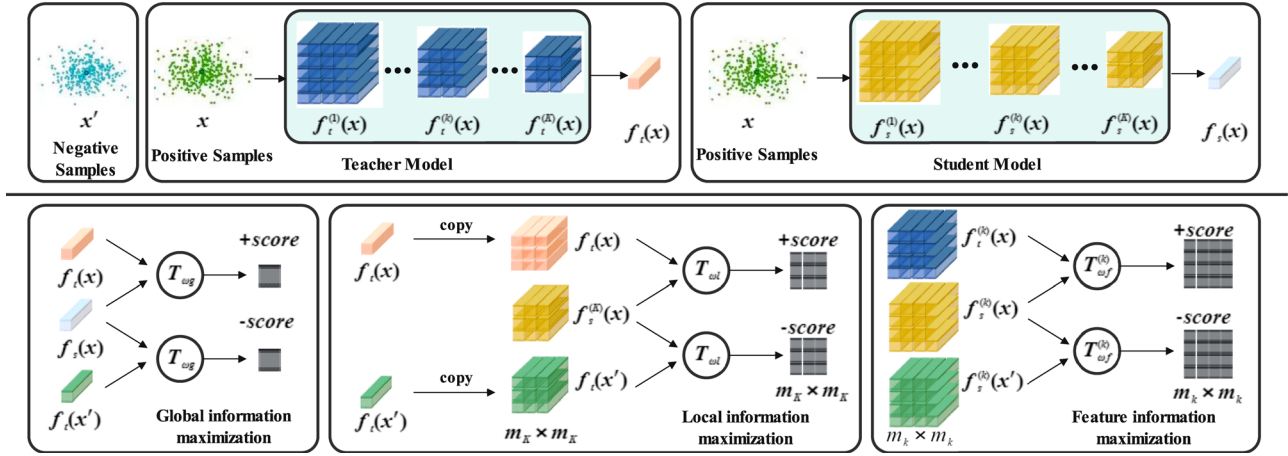


Figure 2. The proposed KF-IDS architecture.

4.2. The proposed KF-IDS

Utilizing mutual information-based knowledge distillation, we develop an enhanced federated learning intrusion detection model tailored to the computational constraints of IoT devices, including limited CPU processing power, restricted memory capacity, and finite battery life, particularly for those operating on networks with local endpoints. Building upon the model introduced in Section 3, our primary objective is to compress the global model while minimizing the impact on the accuracy of individual client models. This approach mitigates the performance drop in model inference that typically occurs when deploying computationally intensive models on resource-limited IoT devices. Our method therefore seeks to achieve an optimal balance between accuracy and complexity.

Figure 2 shows the terminal detection variant of the proposed KF-IDS based on knowledge distillation. We build positive and negative sample pairs based on data sample enhancement and IoT device information, use the trained CF-IDS global model as the teacher model and copy the final feature representation to match the intermediate feature representation of the student model, estimate and maximize the lower bound of mutual information to generate the student model.

4.2.1. Construction of positive and negative sample pairs

The knowledge distillation method based on mutual information relies on positive and negative sample pairs. We fully consider the heterogeneity of IoT devices and use this as a rule for constructing positive and negative samples. For example, given N open datasets, suppose each dataset has a label $\{t_1, t_2, \dots, t_N\}$ that indicates the type of IoT device from which the data originates, such as various types of cameras. Specifically, for a sample x , we assume a

function $t(x)$ that can provide this label. We consider that IoT devices of the same type share common prior knowledge, meaning they operate in similar environments and, therefore, have similar data distributions. Based on this, we construct the following positive and negative samples as follows:

$$t(x_1) = \begin{cases} t(x_2), & \text{if } x_1 \text{ and } x_2 \text{ are positive samples.} \\ t(x_1) \neq t(x_2), & \text{if } x_1 \text{ and } x_2 \text{ are negative samples.} \end{cases} \quad (14)$$

For simplicity in notation, we denote the negative sample corresponding to sample x as x' .

4.2.2. Embedding extraction and mutual information maximization

We denote the already trained model as the teacher model f_t , and the model to be distilled as the student model f_s . Specifically, we denote $f_t^{(k)}$ and $f_s^{(k)}$ as the intermediate representation (or intermediate feature) of the k th layer of the teacher/student model, and $f_t(x)$ and $f_s(x)$ as the logits of the teacher/student model for the sample x .

Consider a set $R = (f_t^{(k)}(x), f_s^{(k)}(x'))$, which contains K pairs of intermediate representations extracted from the network. Each pair in the set R consists of intermediate representations of the same size, where $m_k \times m_k$ represents the size corresponding to the k th pair of intermediate representations in the set. Each position in these two-dimensional intermediate representations corresponds to a specific region in the input. Note that the final representations $f_t(x)$ and $f_s(x)$ are not included in the set R .

Our overall goal is to maximize the mutual information (MI) between the final representations of the frozen teacher network f_t and the student network f_s , thereby encouraging the student to learn richer, more

informative representations. Mutual information quantifies the amount of information obtained about one random variable through another; in this scenario, it measures how much information the student's representation contains about the teacher's representation. Maximizing MI ensures that the student network not only mimics the teacher's output but also captures the underlying patterns and features learned by the teacher, leading to enhanced generalization capabilities.

This goal is achieved using a discriminator function T_ω , where ω represents the trainable parameters of the discriminator. The discriminator's role is pivotal in estimating the mutual information between the teacher and student networks' representations. Additionally, we denote the parameters of f_s as θ . By introducing the discriminator, we transform the problem of maximizing mutual information into an adversarial game, where the student network tries to fool the discriminator by producing representations that are indistinguishable from those generated by the teacher network.

We achieve global mutual information maximization based on the Mutual Information Neural Estimation (MINE) method (Belghazi et al., 2018). MINE provides a scalable and efficient way to estimate and optimize the mutual information between high-dimensional random variables, such as the representations produced by deep neural networks. Specifically, MINE leverages the Donsker-Varadhan dual representation of the Kullback-Leibler divergence to estimate the mutual information. The optimization process involves jointly optimizing both the student's parameters θ and the parameters of the discriminator ω , as follows:

$$\max_{\theta} \min_{\omega} \mathbb{E}_{x \sim p(x)} [T_\omega(f_t(x), f_s(x))] - \log(\mathbb{E}_{x \sim p(x), x' \sim p(x')} [\exp(T_\omega(f_t(x), f_s(x'))))] \quad (15)$$

where $p(x)$ denotes the data distribution, and $p(x')$ represents a marginal distribution used for constructing negative samples. This formulation encourages the student network to generate representations that are close to those of the teacher for real input pairs $(f_t(x), f_s(x))$, while pushing apart the representations for mismatched or negative pairs $(f_t(x), f_s(x'))$. Through iterative training with this objective, the student network gradually learns to produce representations that capture the essential characteristics of the teacher's representations, thereby enhancing its learning capacity and performance.

By incorporating mutual information maximization into the training process, we ensure that the student model not only replicates the outputs of the teacher but also learns a more robust and informative feature space, which is critical for complex tasks such as classification, clustering, and prediction.

Our overall goal is to maximize the mutual information between the final representations of the frozen teacher network f_t and the student network f_s , thereby encouraging the student to learn richer representations. This goal is achieved using a discriminator function T_ω , where ω represents the trainable parameters. Additionally, we denote the parameters of f_s as θ . We achieve global mutual information maximization based on the Mutual Information Neural Estimation (MINE) method (Belghazi et al., 2018). The optimization process involves jointly optimizing both the student's parameters and the parameters of the discriminator, as follows:

$$(\omega, \theta) = \arg \max_{\omega, \theta} I(f_t(x), f_s(x')) \quad (16)$$

where $I(X, Y)$ represents the mutual information of variables X and Y , which can be defined as the Kullback-Leibler (KL) divergence of the product between the joint probability $p(y|x)p(x)$ and the marginal probabilities $p(y)p(x)$:

$$\begin{aligned} I(X, Y) &= \iint p(x, y) \log \frac{p(x, y)}{p(x)p(y)} dx dy \\ &= D_{KL}(p(y|x)p(x) \| p(y)p(x)) \end{aligned} \quad (17)$$

4.2.3. Training strategy

In Section 4.2.1, we constructed two types of positive and negative sample pairs: those based on data augmentation and those based on IoT device type labels. During training, we primarily rely on the positive and negative sample pairs generated through data augmentation, which is controlled by a balancing parameter α . The final loss function is as follows:

$$L_{kd} = L_{kd}^1 + \alpha L_{kd}^2 + \gamma L_{CE}(x, y) \quad (18)$$

where L_{kd}^1 represents the training loss based on the data-augmented samples, and L_{kd}^2 represents the training loss based on the samples labeled by IoT device type. Despite different sample sources, L_{kd}^1 and L_{kd}^2 's input representations are transformed into a comparable feature space, making their loss computations structurally identical. Thus L_{kd}^1 and L_{kd}^2 are computed in the same way:

$$L_{kd}^1 = L_{kd}^2 = I(f_t(x), f_s(x')) + \beta I(f_t(x), f_s(x')) \quad (19)$$

$L_{CE}(x, y)$ is classification loss based on cross-entropy, where y is the label for sample x , and this label is provided by the clients as described in Section 3.

5. Experiment results

5.1. Settings

5.1.1. Dataset

We use the N-BaloT dataset (Meidan et al., 2018) for our IoT traffic experiments, which is an open dataset containing data from nine IoT devices. Seven of these devices have 11 types of traffic: one type of benign traffic and ten types of attack traffic. The remaining two devices have six types of traffic: one benign and five attack types. Each traffic packet includes 115 features extracted from five different time windows.

Specifically, N-BaloT is meticulously crafted for the investigation of botnet attack detection within the realm of Internet of Things (IoT) devices. It encompasses real network traffic data collected from nine commercially available IoT devices, featuring both benign operational traffic and malicious traffic generated under the influence of two notorious botnets, Mirai and BASHLITE. The dataset is designed to capture a comprehensive set of statistical features extracted from various time windows (100 ms, 500 ms, 1.5 s, 10 s, and 1 min), each yielding a 23-dimensional vector that collectively form a single 115-dimensional feature vector per sample. This approach allows researchers to analyze behavioral snapshots indicative of network activity context, facilitating the differentiation between normal operations and attacks such as COMBO, SPAM, and SCAN, which are characteristic of BASHLITE botnet activities.

To ensure the robustness and efficacy of analytical models developed using this dataset, rigorous preprocessing steps are essential. These include data cleaning to address issues like missing values, noise, and outliers through methods such as deletion or imputation with mean or median values; feature selection to identify the most relevant subset of features, thereby enhancing model performance and efficiency; and feature scaling to standardize the range of input variables, preventing any particular feature from disproportionately influencing the outcome due to its scale. Additionally, appropriate data transformations may be applied to meet specific analytical requirements, followed by partitioning the dataset into training, validation, and test sets to facilitate model training, hyperparameter tuning, and final performance evaluation. Through careful analysis and preprocessing of the N-BaloT dataset, researchers can develop more accurate and effective detection mechanisms tailored for securing IoT environments against sophisticated threats.

The N-BaloT dataset holds significant relevance to IoT environments by providing a comprehensive resource for studying and mitigating botnet attacks that specifically target IoT devices. Comprising network traffic data from nine distinct commercial IoT devices, this dataset

captures both benign operational patterns and malicious behaviors induced by the Mirai and BASHLITE botnets. It enables researchers to analyze a wide array of behavioral snapshots through statistical features extracted across multiple time windows, facilitating the identification and differentiation of various attack types such as COMBO, SPAM, and SCAN. Given the increasing ubiquity and vulnerability of IoT devices, the N-BaloT dataset serves as a crucial tool for developing advanced anomaly detection methods, enhancing security measures, and ultimately safeguarding the integrity and functionality of IoT ecosystems against emerging cyber threats.

Before starting the experiments, we allocated the dataset to each client based on the experimental setup, dividing the data according to a Dirichlet distribution $\text{Dir}(\theta)$. As θ increases, the data distribution becomes more independent and identically distributed (IID). As θ approaches zero, the data distribution becomes non-IID.

We then divided the dataset into private, open, and test datasets in proportions of 70%, 20%, and 10%, respectively. The open dataset is unlabeled, while the private dataset is distributed to each client. The divided data has the following characteristics: (1) Each client's private dataset comes exclusively from one device in the original dataset, preserving the original data characteristics. (2) The amount and types of data each client receives vary significantly, resulting in a non-IID data distribution. (3) The private dataset, open dataset, and test dataset for each client have no overlap.

Note that our pre-processing method about the dataset are same as Zhao et al. (2022). Zhao et al. (2022) is already a valuable article on IoT intrusion detection systems for Internet environments. We believe that by aligning the preprocessing approach with Zhao et al. (2022), we can obtain an unbiased performance evaluation.

5.1.2. Soft/Hardware environment

All experiments are implemented on a single NVIDIA RTX 4090 GPU. The software version used is: Pytorch 2.3.0, CUDA 11.8.

5.2. Performance evaluation on CF-IDS

To accurately evaluate the intrusion detection capabilities of the proposed CF-IDS model, we replicated four federated learning intrusion detection methods: FedAvg (McMahan et al., 2017), FedProx (T. Li et al., 2020), Scaffold (Karimireddy et al., 2020), and FedNova (Wang et al., 2020). All methods were tested under consistent conditions, with each algorithm trained for 100 rounds. Specific adjustments were made to adapt the N-BaloT dataset to these methods, ensuring alignment with the experimental criteria.

Table 1. The comparison of accuracy for each federated learning-based IDS.

	CF-IDS	FedAvg	FedProx	Scaffold	FedNova
$\theta \rightarrow 0$	0.7954	0.6808	0.7095	0.7518	0.7654
$\theta = 0.1$	0.8071	0.6538	0.7393	0.6833	0.7011
$\theta = 1$	0.7675	0.5606	0.6097	0.7159	0.7247
$\theta \rightarrow \infty$	0.7577	0.6065	0.6123	0.7033	0.7241
$\theta \rightarrow 0$	0.7954	0.6808	0.7095	0.7518	0.7654

5.2.1. Comparison with different data distribution

This section examines how different data distributions affect various federated learning intrusion detection systems. As described earlier with the Dirichlet distribution, when θ is large, the data distribution exhibits characteristics of independent and identically distributed (IID) data. Conversely, when θ is small, the data distribution shows characteristics of non-IID data. We specifically evaluate four different data distribution scenarios:

- (1) When θ is close to 0, each client receives only one type of network traffic data.
- (2) When $\theta = 0.1$, the data distribution is non-IID, meaning each client receives different types and quantities of network traffic data samples.
- (3) When $\theta = 1$, the data distribution starts to resemble an IID distribution.
- (4) When θ approaches positive infinity, the data distribution is fully IID, with each client receiving the same data type and an equal number of network traffic data samples.

For this analysis, four federated learning intrusion detection methods—FedAvg, FedProx, Scaffold, and FedNova—were used, with a total of 50 clients undergoing 100 rounds of training. The experimental results are shown in Table 1 and Figure 3.

As illustrated in Table 1 and Figure 3, the proposed CF-IDS consistently demonstrates superior accuracy, regardless of whether the data distribution is IID or non-IID. All federated learning-based IDS models perform better with IID data compared to non-IID data. When the local private data of each CF-IDS client becomes more IID, the pseudo labels generated by the central server enhance the training effectiveness of local clients and accelerate model convergence. Consequently, the global model aggregated by the central server captures more comprehensive features and delivers better performance than other FL-based IDS models.

5.2.2. Comparison with the centralized learning-based IDS

In centralized learning-based IDS, all data is stored in a central data agent for training, which poses a risk of

Table 2. The comparison results between federated and centralized learning-based IDS.

	Acc(%)	Pre(%)	Recall(%)	F1(%)
Centralized Learning	82.11	82.85	96.81	89.29
Federated Learning	80.71	82.73	95.21	88.53
Client 1	77.40	79.84	94.16	86.41
Client 2	77.51	80.06	94.07	86.50
Client 3	78.69	80.63	94.93	87.20
Client 4	79.64	81.72	94.95	87.84

data privacy breaches. In contrast, federated learning-based IDS keeps data on local clients, sharing only model parameters during updates, thus providing better privacy protection. To assess the performance of the proposed CF-IDS, we compared its accuracy, precision, recall, and F1 scores with those of a centralized learning-based IDS.

In this experiment, we set the number of clients to 50, with each client undergoing 100 training rounds where $\theta = 0.1$, and randomly selected four clients for performance evaluation. The results are presented in Table 2 and Figure 4.

The comparison shows that the centralized learning-based IDS has a slight edge over the proposed CF-IDS, with 1.40% higher accuracy, 0.12% higher precision, 1.60% higher recall, and 0.76% higher F1 score. These results indicate that centralized learning-based IDS performs better in detection. However, considering the challenge that direct data aggregation is difficult to achieve for the IoT node classifiers in distributed scenarios, the performance loss of CF-IDS is acceptable.

Additionally, when comparing the federated learning-based IDS model to local models trained by individual clients (clients 1 to 4), we see that the accuracy, precision, recall, and F1 scores of the local models are lower than those of the federated learning-based IDS. This demonstrates the practical significance of federated learning for IoT intrusion detection.

5.2.3. Comparison with different ML-based IDS models

To better evaluate the detection performance of the ViT model used in the proposed FL-IDS, we also select 6 different models for comparative experiments, including multi-layer perceptron (MLP), GRU, support vector machine (SVM), CNN, Deep Neural Networks (DNN), RNN. In this experiment, we set the number of clients to 50, the number of client training rounds to 100, and $\theta = 0.1$. The experimental results are shown in the Table 3 and Figure 5.

As shown in Table 3 and Figure 5, the CF-IDS built with the ViT model outperforms other IDS systems that use traditional deep learning or machine learning classifiers in terms of accuracy, precision, recall, and F1 score.

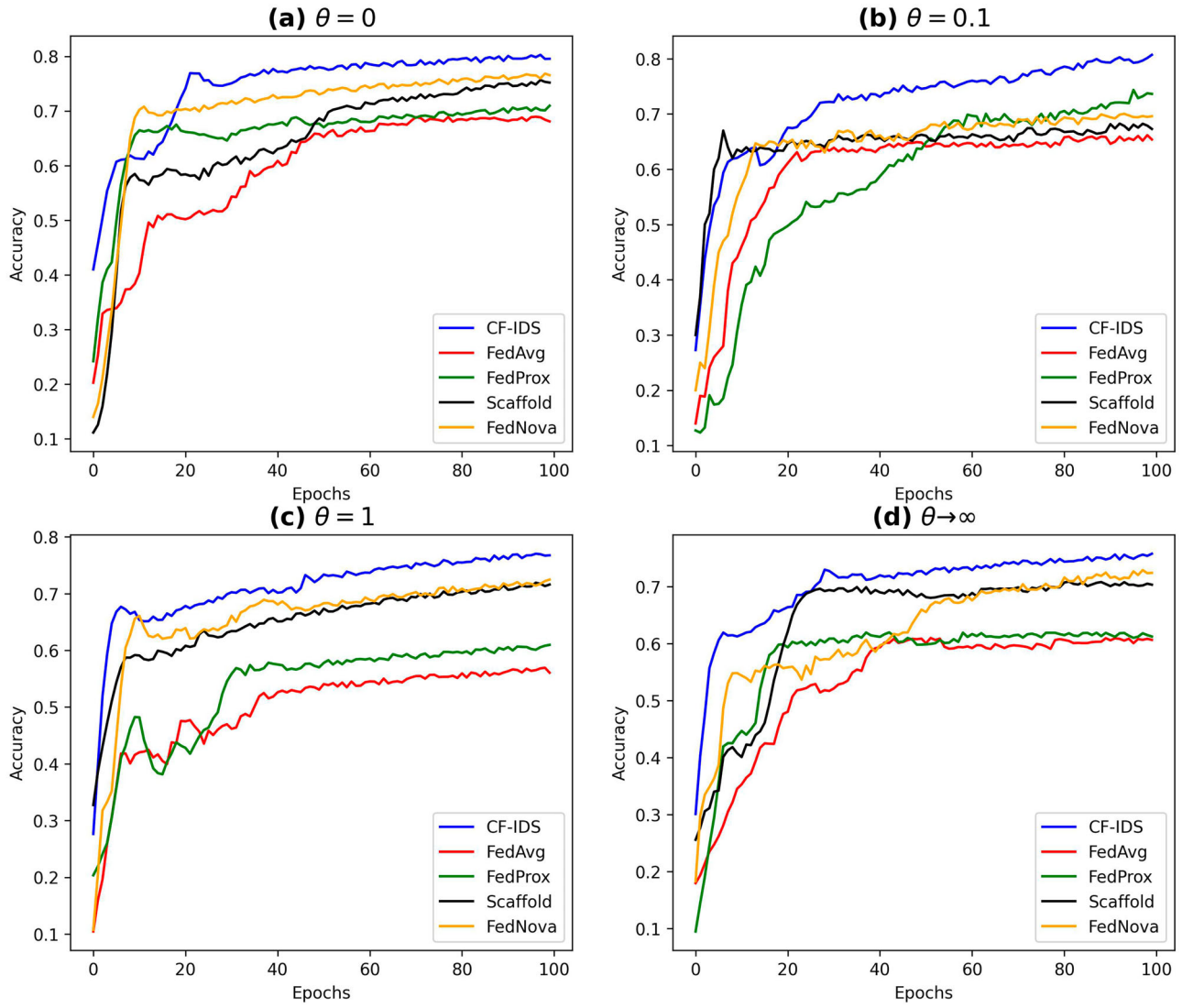


Figure 3. The comparison results of FL-based IDS with different data distribution. (a) single traffic data; (b) non IID data; (c) near IID data; (d) IID data.

Table 3. The comparison results between different DL and ML-based IDS.

	Acc(%)	Pre(%)	Recall(%)	F1(%)
ViT(CF-IDS)	80.71	82.73	95.21	88.53
MLP	71.11	76.19	90.19	82.60
GRU	73.31	77.84	91.22	84.00
SVM	67.40	73.17	88.62	80.16
CNN	70.52	76.62	89.10	82.39
RNN	74.37	78.38	91.95	84.63

This suggests that the proposed CF-IDS with the ViT model has a performance advantage. The system leverages the ViT model's efficient feature extraction capabilities and self-attention mechanism to uncover intrinsic relationships within data samples, allowing for the assignment of more reliable pseudo labels to the training dataset. By incorporating the ViT model, the proposed CF-IDS significantly enhances the accuracy and efficiency

of intrusion detection, while also improving system robustness.

5.2.4. Comparison with different number of clients

Unlike centralized learning, in a federated learning-based intrusion detection system, data remains on each client, and only model parameter updates are shared. However, the number of clients and their computing power still influence the global model's aggregation and updates in the federated learning process. In this section, we examine how varying the number of clients impacts the performance of the federated learning-based IDS model. The training was conducted over 100 rounds with θ set to 0.1, and the number of clients set at 10, 50, 100, and 200. The experimental results are presented in Figure 6 and Table 4.

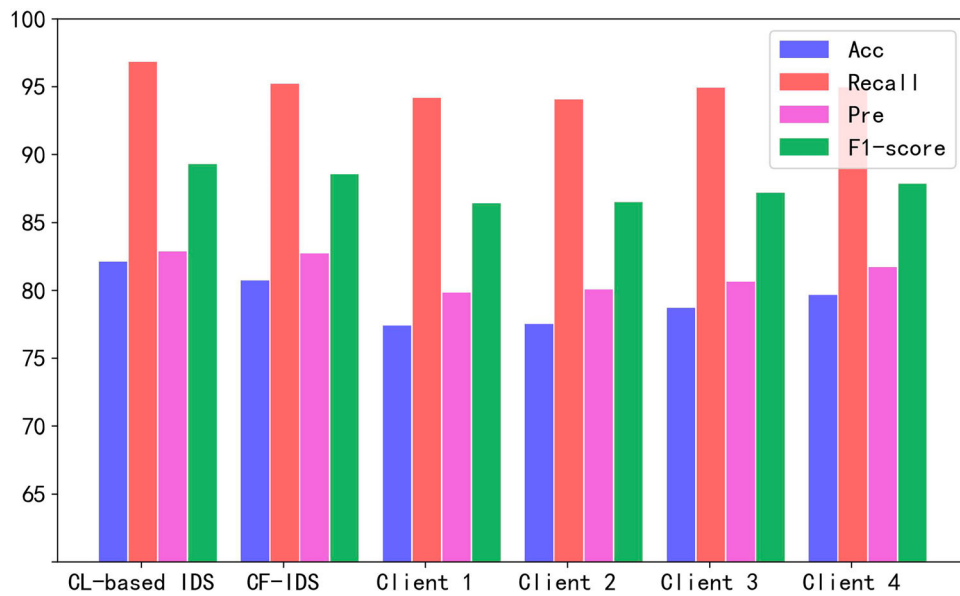


Figure 4. The comparison results between FL-based and CL-based IDS.

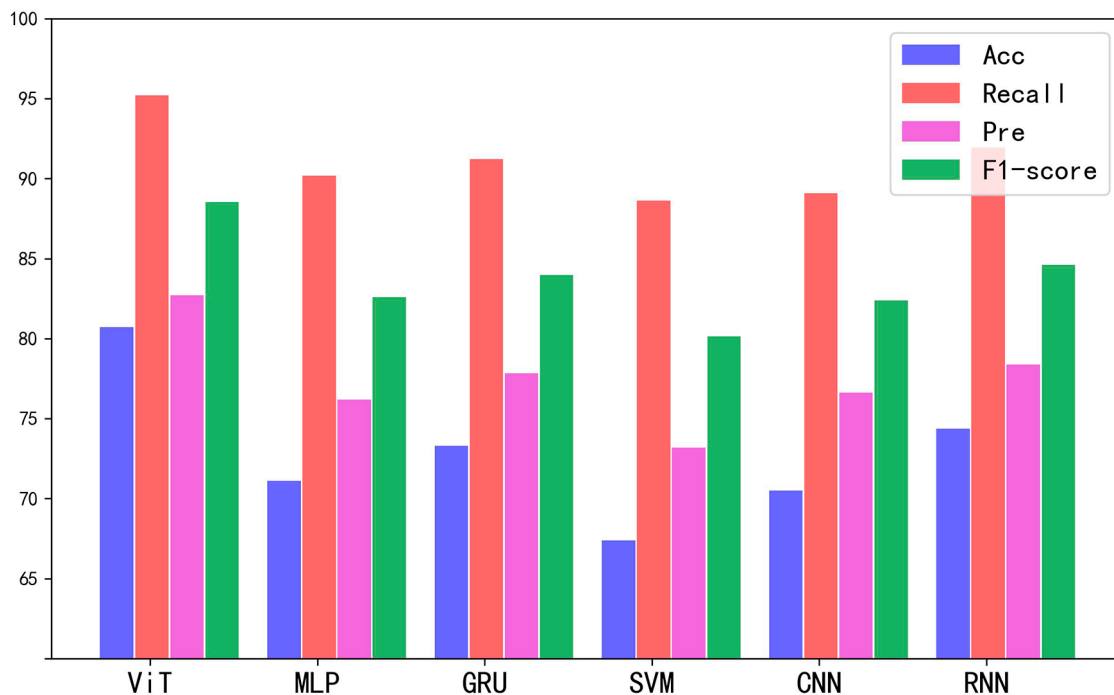


Figure 5. The comparison results between different DL and ML-based IDS.

Table 4. The impact on FL-based IDS with different numbers of clients.

	CF-IDS	FedAvg	FedProx	Scaffold	FedNova
K = 10	0.8188	0.6734	0.7010	0.7628	0.7510
K = 50	0.8071	0.6538	0.7393	0.6833	0.7011
K = 100	0.7547	0.6404	0.6853	0.7180	0.7145
K = 200	0.7256	0.5835	0.6178	0.6667	0.6556

As shown in Figure 6 and Table 4, the accuracy of the intrusion detection system (IDS) decreases as the number of clients participating in federated learning increases.

However, the proposed CF-IDS still maintains an accuracy of 72.56% in the final experiment, outperforming other intrusion detection mechanisms. When the number of clients increased from 100 to 200, the highest accuracy of the FedAvg algorithm dropped to 58.35%, while the accuracies of the FedProx, Scaffold, and FedNova algorithms decreased to 61.78%, 66.67%, and 65.56%, respectively. This decline in accuracy is due to the reduced number of samples each client receives when the total amount of data remains constant but the number of clients increases. As a result, the local prediction models

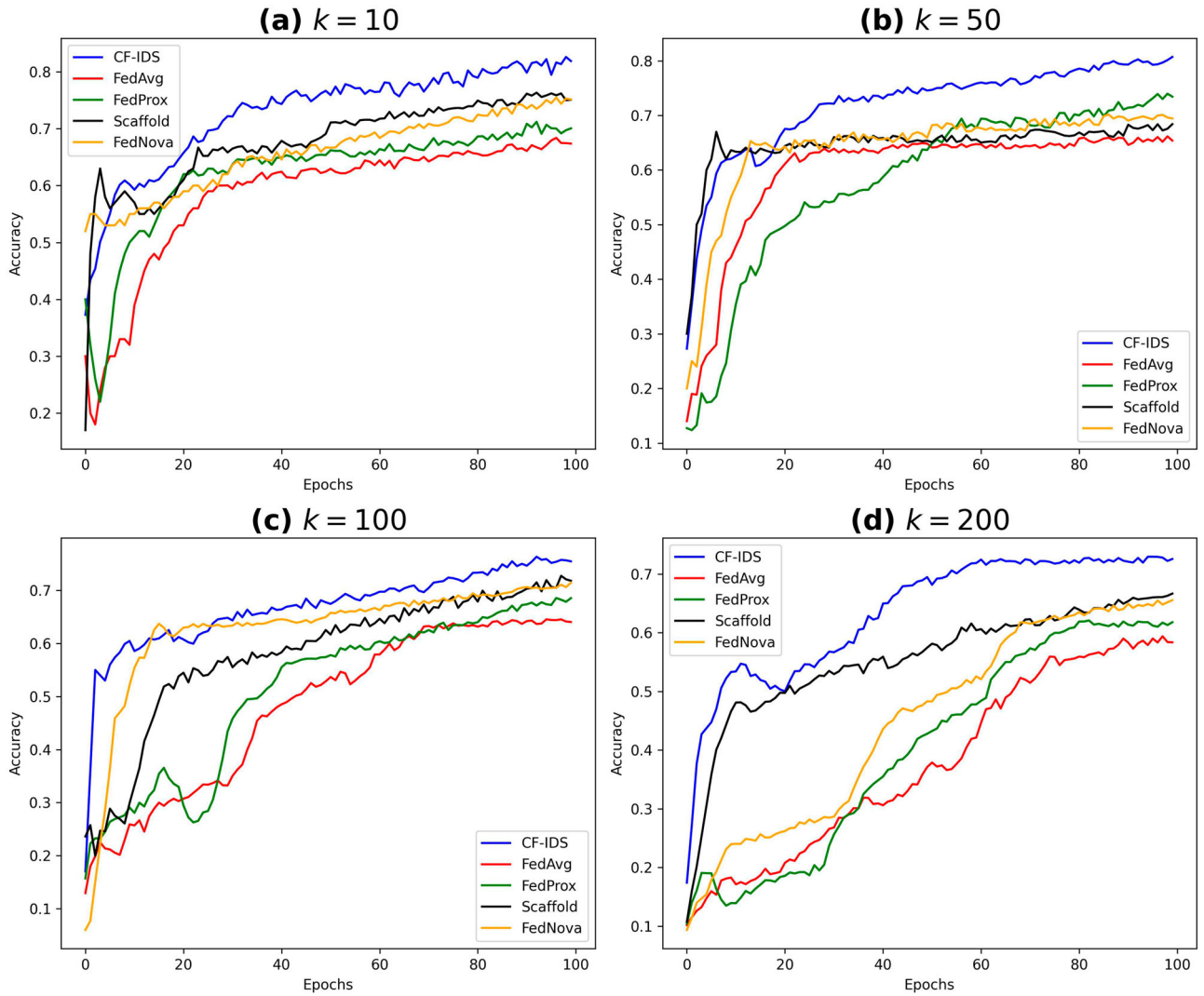


Figure 6. The comparison results of FL-based IDS with different numbers of clients.

struggle to effectively extract the characteristics of IoT traffic data, leading to decreased accuracy.

Unlike other federated learning-based IDS schemes, the proposed CF-IDS employs contrastive learning and the K-means clustering algorithm to generate pseudo labels, helping clients train their models more effectively. This approach mitigates the impact of having fewer traffic samples during the retraining of the FL-based IDS classifier model. Consequently, the accuracy of the proposed CF-IDS model remains high even as the number of clients decreases, demonstrating its feasibility and practical significance in real-world IoT intrusion detection applications.

5.2.5. Ablation experiment

In order to better verify the effectiveness of the method proposed in this study, we set up three groups of ablation experiments, namely, federated learning without pseudo labels, federated learning without discriminators, and

one-class classifier experiments. The fourth group is the training scheme proposed in this paper that includes pseudo labels and discriminators. The number of training rounds is fixed to 100 rounds, θ is 0.1, and the number of clients is 50. The experimental results are shown in Figure 7 and Table 5.

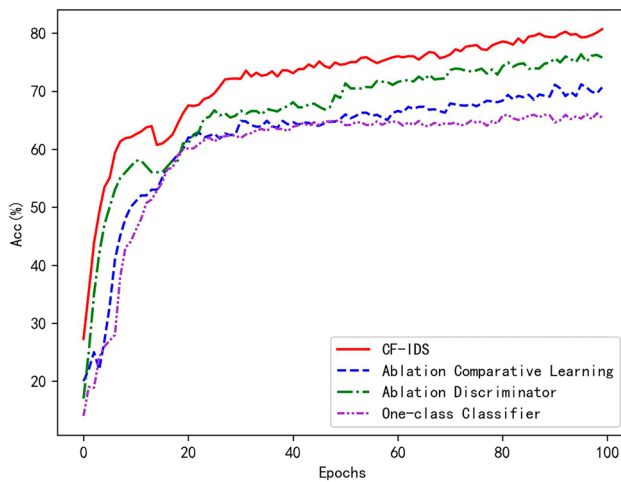
Obviously, ablation contrastive learning and discriminator will significantly affect the detection model performance. Specifically, after removing the client discriminator and only retaining the contrastive learning module, the convergence speed of the model has decreased, which will also affect the highest accuracy at the end of training. At the same time, if we remove the contrastive learning module (no pseudo label generation), the convergence speed of the CF-IDS will be greatly reduced, and the accuracy at the end of training will also be reduced, and will be lower than the accuracy of the federated learning model that only ablates the discriminator. This shows that the contrastive learning module plays a more

Table 5. Accuracy of different rounds of training in ablation experiments.

	CL	Discriminator	10	25	50	100
CF-IDS	✓	✓	0.6205	0.6894	0.7456	0.8071
Ablation contrastive learning	✓	×	0.5015	0.6231	0.6519	0.7065
Ablation Discriminator	×	✓	0.5709	0.6555	0.6885	0.7581
One-class Classifier	×	×	0.4413	0.6227	0.6483	0.6538

Table 6. Computation time and percentage comparison of teacher model with different compression multiplicity KF-IDS model.

	Transformer	Transformer/2	Transformer/3	Transformer/4	LSTM
Time(s)	9.461	5.032	3.817	2.759	2.656
Teacher Model Percentage(%)		65.3	71.3	77.4	78.1
Student Model Percentage(%)		34.7	28.7	22.6	21.9

**Figure 7.** The accuracy of different rounds in ablation experiments.

important role in accelerating model convergence and improving model accuracy.

5.3. Performance evaluation on KF-IDS

To assess the detection performance of the proposed KF-IDS in practical applications, we conducted various experiments, including evaluating the model's performance under different compression ratios, examining the impact of different data distributions on the model, and analysing the effect of varying the number of clients. The experimental environment used was consistent with that described in Section 5.1.

5.3.1. Comparison of time consumption and memory usage

To further demonstrate the impact of knowledge distillation on the model and to evaluate the role of the KF-IDS model in IoT environments, we compared the computation time of the teacher model with that of the KF-IDS model. Additionally, we compared the percentage of

computation consumption before and after distillation. The results are shown in Table 6.

Compared to the teacher model, the KF-IDS model after knowledge distillation significantly reduces computation time, and as the compression ratio increases, the computation time decreases further. When the student model uses Transformer/2, the computation time is only 34.7% of the total time cost. Similarly, when the student model uses Transformer/3, Transformer/4, and LSTM models, the computation time is 28.7%, 22.6%, and 21.9%, respectively.

Furthermore, the memory usage of a model directly reflects its essential resources requirements. Models with lower memory consumption can be more widely deployed in common IoT scenarios, making it practically significant to evaluate the memory usage of the KF-IDS model. The results are shown in Table 7.

As shown in Table 7, the memory usage of the KF-IDS student model after knowledge distillation is significantly reduced compared to the teacher model, with student models having higher compression ratios consuming even less memory. The memory consumption of the student model using Transformer/2 is only 34.1% of the total memory usage. Similarly, the memory consumption of the student models using Transformer/3, Transformer/4, and LSTM is 25.4%, 19.8%, and 19.4% respectively. The memory usage of the student models has been greatly reduced compared to the teacher model.

5.3.2. Accuracy at different compression ratios

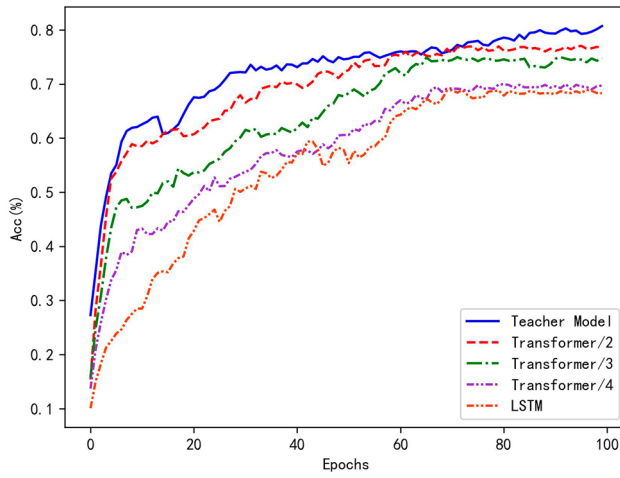
In this section, we aim to evaluate the impact of different compression ratios on the KF-IDS model. For this training, the number of clients is fixed at 50, with 100 training rounds, and θ is set to 0.1. The teacher model is compressed by 2x, 3x, and 4x to generate the student models. The student models used are the Transformer models compressed by 2x, 3x, and 4x (denoted as Transformer/2, Transformer/3, and Transformer/4) and the LSTM model. The comparison results of the KF-IDS model under different compression ratios are shown in Figure 8 and Table 8.

Table 7. Memory usage and percentage comparison of teacher model with different compression multiplicity KF-IDS model.

	Transformer	Transformer/2	Transformer/3	Transformer/4	LSTM
Memory Usage(MB)	47.89	24.76	16.34	11.79	11.55
Teacher Model Percentage(%)		65.9	74.6	80.2	80.6
Student Model Percentage(%)		34.1	25.4	19.8	19.4

Table 8. Accuracy of each model with different compression ratios.

Teacher network	Transformer	Transformer	Transformer	Transformer
Student network	Transformer/2	Transformer/3	Transformer/4	LSTM
Teacher's Accuracy	80.71	80.71	80.71	80.71
Student's Accuracy	74.82	72.66	67.12	65.79
KF-IDS	76.73	74.69	69.71	68.29

**Figure 8.** Accuracy of the KF-IDS at different compression ratios.

From Table 8, it can be seen that the KF-IDS intrusion detection model consistently outperforms the student model. When the teacher model is compressed by a factor of 2, KF-IDS achieves an accuracy 1.91% higher than the student model. With a $3 \times$ compression, KF-IDS surpasses the student model by 2.03%, and with a $4 \times$ compression, it outperforms the student model by 2.59%. When the student model uses an LSTM, KF-IDS achieves 2.50% higher accuracy than the student model.

To better evaluate the intrusion detection performance of the KF-IDS model, we also reproduce three representative intrusion detection models for comparison: Isolation Forest (IF), BeatGAN and OmniAnomaly, and compare the accuracy of KF-IDS with the other intrusion detection models in different compression ratios. The accuracy results are shown in Table 9.

- *Isolation Forest (IF, Liu et al., 2008)*: One of the most representative machine learning-based anomaly detection methods. This method detects anomalies by marking samples that are sparsely distributed and far from high-density clusters as anomalies.

- *BeatGAN (Zhou et al., 2019)*: An anomaly detection method based on the GAN model. It employs an autoencoder structure and a discriminator for adversarial regularization to mitigate model overfitting. The model primarily detects anomalous time-series samples by comparing the input time series with its reconstructed counterpart generated in reverse.
- *OmniAnomaly (Su et al., 2019)*: An anomaly detection method based on the VAE model. It leverages techniques such as stochastic variable connections and planar normalizing flows to address the temporal dependencies in multivariate time series.

As shown in Table 9, the KF-IDS intrusion detection model generally outperforms other intrusion detection methods in most cases. Additionally, as seen in Figure 8, the student models generated by different compression ratios for the KF-IDS model are overall less accurate than the teacher model. This indicates that knowledge distillation still results in accuracy loss, and the greater the compression ratio, the greater the accuracy loss. When the KF-IDS model's student models use Transformer/2, Transformer/3, Transformer/4, and LSTM models, the accuracy losses are 3.98%, 6.02%, 11.00%, and 12.42%, respectively.

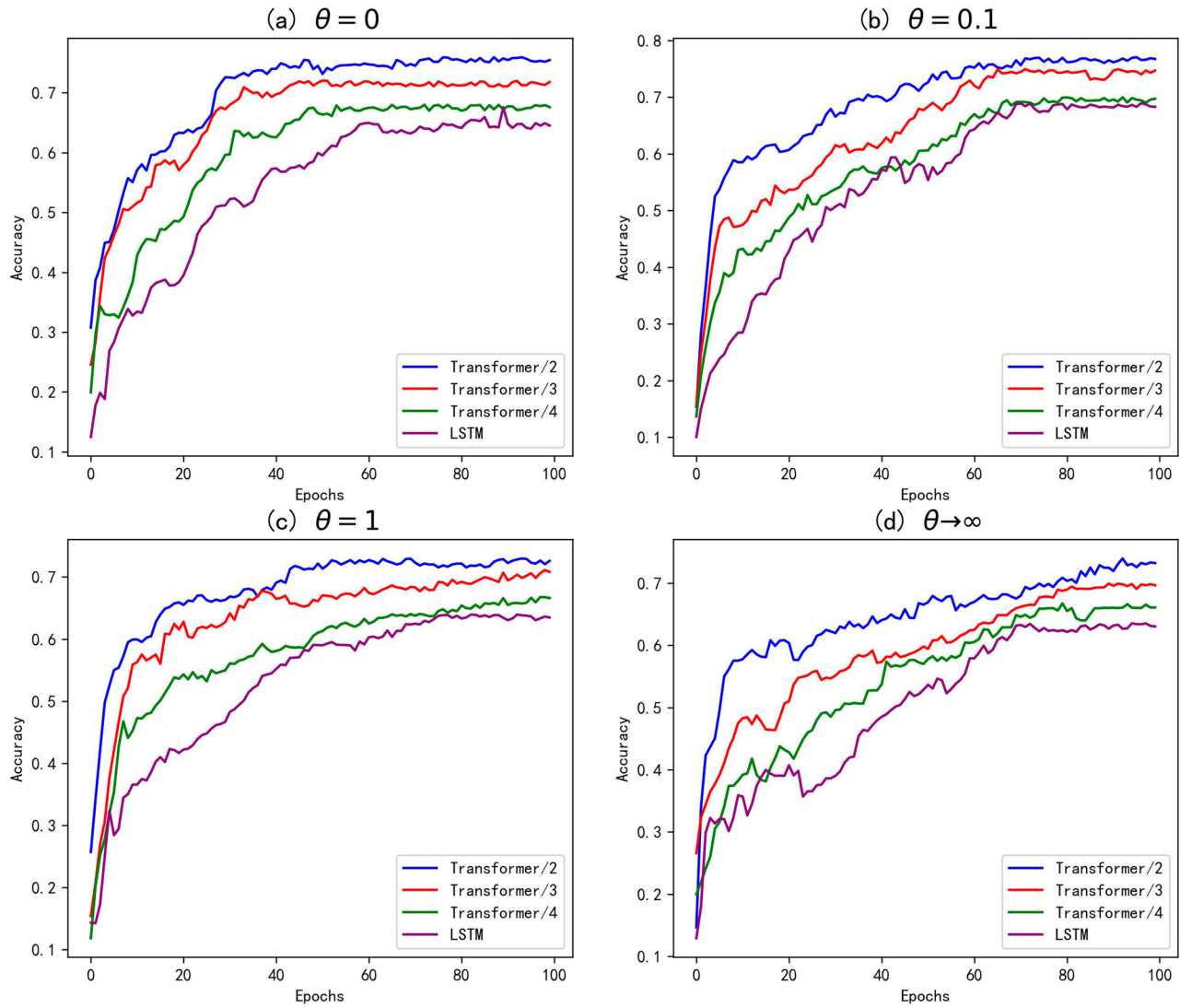
5.3.3. Comparison with different data distribution

In this experiment, the Dirichlet function's θ values are set to 0, 0.1, 1, and ∞ , representing single traffic data, non-independent and identically distributed (non-IID) data, near-independent and identically distributed (near-IID) data, and fully independent and identically distributed (IID) data, respectively. The data distribution settings are the same as those in Section 4.1.1. Additionally, the number of clients is fixed at 50, with each client undergoing 100 training rounds. The experimental results are shown in the Figure 9 and Table 10.

As shown in Figure 9 and Table 10, the KF-IDS model maintains a relatively high accuracy regardless of whether

Table 9. Comparison of the accuracy of KF-IDS and various IDS with different compression ratios.

	IF	BeatGAN	OmniAnomaly	Transformer/2	Transformer/3	Transformer/4
Acc	0.7438	0.759	0.7638	0.7673	0.7469	0.6971

**Figure 9.** KF-IDS accuracy for different data distributions. (a) single traffic data; (b) non IID data; (c) near IID data; (d) IID data.**Table 10.** KF-IDS accuracy for different data distributions. (a) single traffic data; (b) non IID data; (c) near IID data; (d) IID data.

	Transformer/2	Transformer/3	Transformer/4	LSTM
$\theta \rightarrow 0$	0.7547	0.7181	0.6761	0.6453
$\theta = 0.1$	0.7673	0.7469	0.6971	0.6829
$\theta = 1$	0.7258	0.7081	0.6661	0.6346
$\theta \rightarrow \infty$	0.7123	0.6765	0.6410	0.6104

the data distribution is independent and identically distributed (IID) or non-independent and identically distributed (non-IID). However, as the compression ratio of the student model increases, the accuracy loss of the

KF-IDS model also increases, leading to a decrease in accuracy.

When $\theta = 0.1$ (non-IID data), the accuracy loss for the Transformer/2 student model is 3.98%, and for the LSTM student model, the accuracy loss compared to the teacher model is 12.42%. When θ approaches $+\infty$ (IID data), the accuracy loss for the Transformer/2 student model is 9.48%, and for the LSTM student model, the accuracy loss compared to the teacher model is 19.67%. This indicates that in the KF-IDS model, the higher the compression ratio of the student model during knowledge distillation by the central client, the greater the accuracy loss when

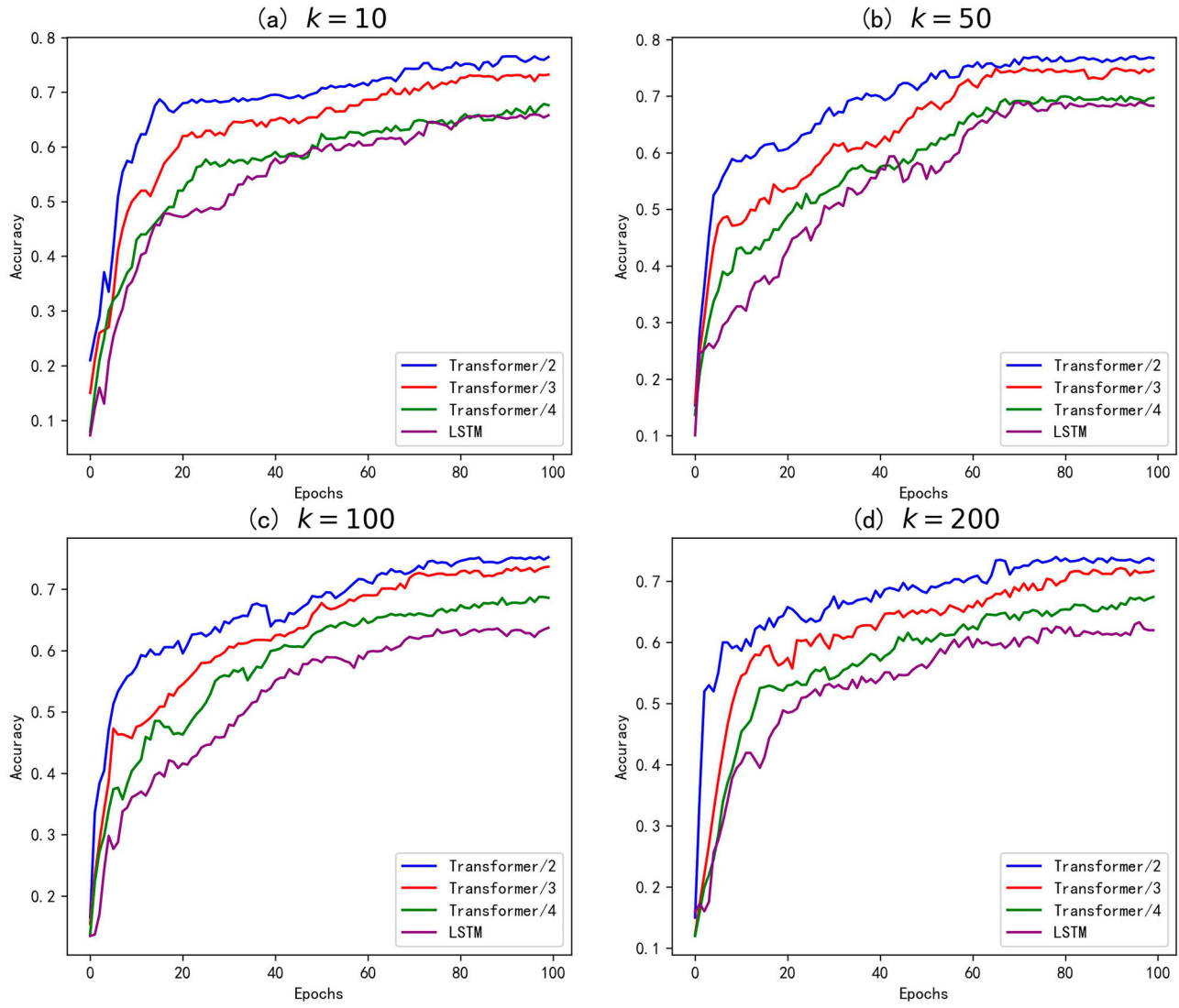


Figure 10. Impact of different number of clients on KF-IDS.

the student model learns from the teacher model. Additionally, as the distribution of local private data on the clients becomes increasingly IID, the accuracy loss of the compressed student model after knowledge distillation becomes more significant.

5.3.4. Comparison with different number of clients

This section verifies the impact of different numbers of clients on the KF-IDS model. The number of training rounds is fixed to 100, θ is 0.1, and the number of clients is set to 10, 50, 100, and 200. The results are shown in Figure 10 and Table 11.

As shown in Figure 10 and Table 11, the number of clients involved in building the KF-IDS system impacts the training of the federated learning model to some extent. However, the compressed KF-IDS model consistently maintains an accuracy of approximately 67.44% to 76.45% across different settings. Additionally, as the

Table 11. KF-IDS accuracy for different number of clients.

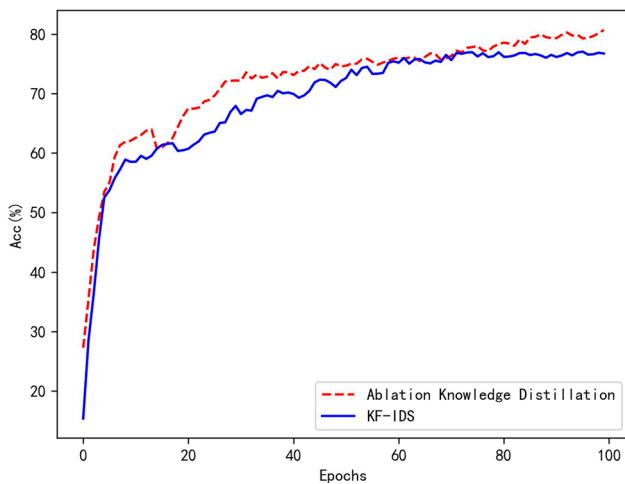
	Transformer/2	Transformer/3	Transformer/4	LSTM
$K = 10$	0.7645	0.7321	0.6762	0.6578
$K = 50$	0.7673	0.7469	0.6971	0.6829
$K = 100$	0.7524	0.7367	0.6861	0.6373
$K = 200$	0.7346	0.7172	0.6744	0.6203

student model undergoes higher compression through knowledge distillation, the accuracy loss increases, resulting in an overall accuracy reduction of 0.23% to 5.43% compared to the uncompressed model.

Specifically, when the number of clients using the Transformer/2 student model is 10, 50, 100, and 200, the accuracy of the KF-IDS model is 76.45%, 76.73%, 75.24%, and 73.46%, respectively, showing a maximum and minimum accuracy difference of 3.27%. In contrast, the LSTM student model exhibits an accuracy of 65.78%, 68.29%, 63.73%, and 62.03% under the same

Table 12. Accuracy of different rounds of training in ablation experiments.

	Knowledge Distillation	10	25	50	100
KF-IDS	✓	0.5851	0.6339	0.7211	0.7673
Ablation Distillation	×	0.6205	0.6894	0.7456	0.8071

**Figure 11.** Accuracy of different rounds of training in ablation experiments.

client numbers, with a maximum and minimum accuracy difference of 6.26%. This indicates that as the number of clients increases, higher model compression ratios have a more significant impact on the model's overall performance. Although student models with lower compression ratios also experience some accuracy loss, their training time and memory usage are significantly better than those of the teacher model, aligning with the IoT environment's requirements for high efficiency and low memory usage.

5.3.5. Ablation experiment

To better verify the feasibility of the proposed KF-IDS model in the IoT environment, we set up a set of ablation experiments, eliminated the knowledge distillation method, observed the model accuracy, and evaluated the effectiveness of knowledge distillation. The number of training rounds was fixed to 100 rounds, θ was 0.1, and the number of clients was 50. The experimental results are shown in Table 12 and Figure 11.

We observed that after the knowledge distillation method was eliminated, the final accuracy of the model increased by 3.98%, indicating that the compressed model would affect the aggregation and accuracy of the model. However, knowledge distillation has greater advantages in computing speed and memory consumption. Considering the uneven performance of IoT devices in the IoT environment, the decreased accuracy of the KF-IDS intrusion detection model is still within an acceptable range, and the superior memory consumption

and computing speed are undoubtedly more suitable for the IoT environment.

6. Discussion

The proposed CF-IDS and its subsequent lightweight version KF-IDS offer a novel approach to addressing the challenges of intrusion detection in the diverse and resource-constrained IoT landscape. CF-IDS tackles the critical issues of data privacy and non-IID data distribution, which are inherent in IoT environments, by leveraging a federated learning framework. The use of contrastive learning on a public dataset to generate pseudo-labels is a key innovation. This strategy allows for better initialization of client models, effectively mitigating the cold-start problem often encountered in federated learning, especially when dealing with highly heterogeneous data from various IoT devices. By aligning the initial knowledge across clients, CF-IDS facilitates faster convergence and improved overall performance in non-IID scenarios, as evidenced by its reported accuracy of 80.71%, outperforming methods like FedNova.

The integration of a discriminator within each client's local training process further refines the quality of pseudo-labels by identifying and specifically labeling unfamiliar traffic. This semi-supervised approach allows the system to learn from both labeled private data and unlabeled public data, enhancing its adaptability to unseen attack patterns. The Shapley value-based voting mechanism introduced during the aggregation phase is another significant contribution. By weighting each client's contribution based on its historical performance and impact on the overall loss, this method moves beyond simple averaging of model parameters, leading to a more robust and accurate global model. This approach is particularly beneficial in scenarios where some clients might have more reliable or representative data than others.

KF-IDS, as a distilled version of CF-IDS, directly addresses the resource limitations of IoT end devices. By employing knowledge distillation and mutual information theory, KF-IDS successfully compresses the knowledge learned by the more complex CF-IDS model into lightweight models suitable for local deployment. The experimental results, showing a significant reduction in time consumption (46.81%) and memory usage (48.30%) while maintaining comparable accuracy, highlight the practical viability of KF-IDS for real-world IoT

deployments. The comparison with deploying on affordable devices like NVIDIA Jetson Nano versus high-end GPUs like RTX 4090 or Tesla A100 further emphasizes the cost-effectiveness and accessibility of KF-IDS for widespread adoption in IoT security.

However, it is important to consider potential limitations. The performance of CF-IDS relies on the quality and relevance of the public dataset used for pseudo-label generation. If the public data does not adequately represent the types of traffic and attack patterns encountered by the client devices, the effectiveness of the initial model alignment might be limited. Furthermore, the choice of contrastive learning techniques, clustering algorithms, and the number of clusters can significantly impact the quality of the pseudo-labels and, consequently, the overall performance. Similarly, the effectiveness of knowledge distillation in KF-IDS depends on the careful design of the student model architecture and the distillation strategy. Future work could explore the sensitivity of the proposed methods to these factors and investigate adaptive mechanisms for selecting and utilizing public data. Additionally, the computational overhead of calculating Shapley values for a large number of clients could be a concern, and exploring more efficient approximation methods might be necessary for scalability.

Despite these potential limitations, CF-IDS and KF-IDS represent a significant step forward in developing privacy-preserving, robust, and resource-efficient intrusion detection systems for the Internet of Things. The integration of contrastive learning, pseudo-labeling, Shapley value-based aggregation, and knowledge distillation provides a comprehensive framework for addressing the unique challenges of IoT security.

7. Conclusion

This paper introduced CF-IDS, a novel federated learning-based intrusion detection system for cloud environments, and KF-IDS, its lightweight counterpart for on-device deployment. By addressing the non-IID nature of IoT data and cold-start issues through contrastive learning and clustering for pseudo-label generation, CF-IDS achieves significant improvements in initial model alignment and accuracy. The integration of a Shapley value-based voting mechanism further enhances robustness and accuracy in global model aggregation.

KF-IDS builds on CF-IDS by employing knowledge distillation and mutual information theory to suit resource-constrained devices, showing notable reductions in time and memory usage while maintaining high accuracy. These advancements make CF-IDS and KF-IDS pivotal contributions to IoT security, effectively tackling challenges like data privacy, non-IID data, and resource limitations.

Looking ahead, future work could explore advanced contrastive learning techniques and adaptive clustering to improve CF-IDS's pseudo-label quality and model alignment. For KF-IDS, investigating tailored knowledge distillation architectures and loss functions can yield more efficient models. Additionally, enhancing both models' resilience against adversarial attacks and optimizing Shapley value calculations for scalability are critical areas for future research. These efforts will ensure that CF-IDS and KF-IDS continue to evolve, providing robust security solutions for an increasingly interconnected world.

Author contributions

Methodology, L.M.; validation, J.H.; writing – original draft preparation, K.L., L.M.; writing – review and editing, D.W., L.Y., and L.M.; visualization, Z.L. All authors have read and agreed to the published version of the manuscript.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Data availability statement

The datasets can be found in the following repositories: <https://www.kaggle.com/datasets/mkashifn/nbaiot-dataset>, accessed on 18 July 2024, reference Meidan et al. (2018).

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Appendix. Abbreviations

Table A1 lists the abbreviations and their corresponding definitions used throughout this paper for clarity and consistency.

Table A1. Abbreviations and their definitions in this work.

Abbreviation	Definition
IDS	Intrusion Detection System
IID	Independent Identical Distribution
near-IID	Near Independent Identical Distribution
non-IID	Non Independent Identical Distribution
CF-IDS	The Proposed Contrastive Learning based Federated IDS
KF-IDS	The Knowledge Distillation-based Version of CF-IDS
DoS	Denial-of-Service Attacks
DDoS	Distributed Denial-of-Service
CNN	Convolution Neural Networks
LSTM	Long Short-Term Memory
WDLSTM	Weighted Long Short-Term Memory
BiLSTM	Bidirectional Long Short-Term Memory
SSFL	Semi-Supervised Federated Learning
ViT	Vision Transformer
MI	Mutual Information
MINE	Mutual Information Neural Estimation
KL	Kullback-Leibler
Dir(θ)	Dirichlet Distribution
MLP	Multilayer Perceptron
GRU	Gated Recurrent Unit
SVM	Support Vector Machine
DNN	Deep Neural Networks
RNN	Recurrent Neural Networks
IF	Isolation Forest
Mirai	A Type of Malware Used in IoT Botnet Attacks
BASHLIGHT	A Botnet Malware Similar to Mirai
COMBO	A Credential Stuffing Attack Using Username-Password Pairs
SPAM	Unsolicited or Malicious Network Traffic
SCAN	Network Scanning Activity for Vulnerability Discovery
FedAvg	Federated Averaging Algorithm
FedProx	Federated Optimization with Proximal Terms
Scaffold	Federated Learning with Control Variates
FedNova	Federated Normalized Averaging Algorithm
BeatGAN	GAN-based Time-Series Anomaly Detection Model
OmniAnomaly	Variational Autoencoder-based Time-Series Anomaly Detection Model